

Education

Master of Science, Computer Science University of California, Santa Cruz - Specialized in Machine Learning and Computer Vision - GPA: 4.00 (Transcript)	Sept. 2024 to Dec. 2025
Bachelor of Science, Computer Science University of California, Santa Cruz GPA: 3.96 (Transcript)	Sept. 2020 to May 2024

Work Experience

InnBlock , Backend Engineer, <i>Fulltime (40h/week)</i> Display case startup where I'm building backend API powering storefront and product detail pages, supporting product listings, pricing, and inventory data. Designing REST APIs to connect frontend UI with product and user data, and implementing server-side logic for product retrieval, filtering, and page rendering	CA, USA March 2026 to Current
NTT Research PHI Lab , Intern, <i>Fulltime (40h/week) June - Oct, Part-time (15h/week) Oct - Dec</i> Research intern studying application of using graph neural networks for solving the quadratic assignment problem (QAP), SAT solver selection, and hyperparameter inference for Ising solvers. Closely worked with mentoring researcher to understand state of the art models and experiment with methods, their efficacies, and implementation to the problem at hand.	CA, USA June 2024 to December 2024
University of California Santa Cruz , Tutor and Grader, <i>Part-time (15h/week)</i> Assisted computer science students on specific projects, closely reviewing materials and assignments to offer targeted help. Courses: Analysis of Algorithms, Computer Architecture, and Computational Models.	CA, USA Sep 2022 to Mar 2024

Projects

Live Text Detector Developed two blur robust real time live video text detection models to aid the visually impaired, achieving over 50% relative F1 improvement in live video over the baseline DBNet model under heavy motion blur.
Efficient Multi-Agent LLM Orchestration for Optimization Developed and evaluated alternative orchestration policies for a multi agent LLM system that converts natural language optimization problems into executable integer programming code. Replaced a costly conductor LLM with rule based and reinforcement learning policies to reduce token usage while maintaining performance.
PEFT LLaVa-Med for X-Ray Disease Classification Finetuned the open-ended multimodal LLaVA-Med model using parameter-efficient methods to detect pneumothorax and multiclass diseases from chest X-rays, achieving a 25%+ absolute improvement in F1-score over zero-shot baselines on the NIH ChestX-ray14 dataset.
GNN SAT Solver Selector Designed and trained a graph neural network to predict the optimal SAT solver for a given Boolean satisfiability instance, optimizing solver selection efficiency across large CNF datasets. Achieving a cumulative runtime between all chosen solvers of 20% the best performing solo model.

Skills

Languages Python (6 Yrs), C (5 Yrs), C++ (5 Yrs), SQL (2 Yrs)	Machine Learning CNNs, Vision Transformers, GNNs, Diffusion Models, NLP, LLMs	Master's Classes Network Security, Distributed Systems, Advanced Computer Vision, Advanced Machine Learning, Artificial Intelligence, Computer Architecture, Algorithm Analysis
ML Libraries & Frameworks Azure AI Foundry, PyTorch, TensorFlow, LLM API, HuggingFace, scikit-learn, NumPy, SciPy, pandas, etc	Systems / Tools Git, AWS, Linux, LaTeX, Docker	

Publications

- Jiang, E., Leleu, T., Reifenstein, S., & Doppalapudi, M. "Graph Neural Networks for Hyperparameter Inference in Ising Solvers." Optimization for ML (OPT) Workshop at NeurIPS 2024, 2024.